Quantum Query Algorithms for the Constructive Diagonal Ramsey Theorem

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Abstract

The constructive diagonal Ramsey problem asks, given adjacency-oracle access to an N-vertex graph, for a clique or independent set of the order guaranteed by Ramsey's theorem. We give a bounded-error quantum algorithm that, for every $K \geq 2$ and $N \geq 4^{K-1}$, finds and verifies a homogeneous K-set using

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

edge queries with failure probability at most η. At the Ramsey scale $N = 2^n$, this yields a homogeneous set of order $\lfloor n/2 \rfloor + 1$ using $O(\sqrt{N} \log N \log(\log N/\eta))$ queries, improving on the $O(N)$ queries of the explicit Erdős–Szekeres recursion and giving, to our knowledge, the first sublinear worst-case algorithm for the Ramsey relation, whose query complexity was raised as a question by Jain, Li, Robere, and Xun (FOCS 2024). The bound also shows that the $N^{1-o(1)}$ quantum lower bound reported there cannot hold at its stated parameters; we show that the cited ingredients compose to an $\Omega(N^{1/12})$ lower bound instead.

The algorithm runs the constructive recursion over implicit candidate sets. Each set is represented by a short conjunction of adjacency constraints and sampled using capped unknown-solution quantum search, and a scale-aware concentration schedule balances estimation accuracy against the increasing cost of sampling deeper sets. We complement the upper bound with an $\Omega(N^{1-1/\sqrt{2}})$ randomized lower bound, transported from the random-Painter analysis of online Ramsey numbers, which holds on the uniform distribution $G(N, 1/2)$. On that distribution a greedy quantum search uses only $\widetilde{O}(N^{1/4})$ queries, giving a provable polynomial quantum speedup for Ramsey search on random graphs. We also give an estimation-free size-biased recursion and extend it to every fixed number of edge colours.

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1 Introduction

Ramsey's theorem guarantees that every graph on $N \geq 4^{K-1}$ vertices contains a clique or an independent set of order K. Its classical constructive proof [8] repeatedly chooses a pivot and restricts the candidate set to one of the pivot's two colour neighbourhoods; truncating each retained neighbourhood to half its current size, it finds a homogeneous K-set with $O(N)$ adjacency queries. The corresponding total search problem, Ramsey, has been studied in proof complexity and in TFNP [15, 16, 14, 20, 13]: the input

model	input	lower bound		upper bound	
randomized	worst case	$\Omega(N^{1-1/\sqrt{2}})$	[Proposition 5.2]	$O(N)$	[8]
randomized	$G(N,1/2)$	$\Omega(N^{1-1/\sqrt{2}})$	[Proposition 5.2]	$\widetilde{O}(N^{1/2})$	[Remark 5.5]
quantum	worst case	$\Omega(N^{1/12})$	[Proposition 6.1]	$\widetilde{O}(N^{1/2})$	[Corollary 1.2]
quantum	$G(N,1/2)$	open		$\widetilde{O}(N^{1/4})$	[Corollary 5.4]

Table 1 Bounded-error query complexity of finding a homogeneous set of order $\lfloor n/2 \rfloor + 1$ in a graph on $N = 2^n$ vertices, in the worst case and on the uniform distribution $G(N,1/2)$; here $1 - 1/\sqrt{2} \approx 0.2929$. The randomized lower bound is the random-Painter estimate of [5] for online Ramsey numbers, previously cited only for deterministic algorithms; the quantum lower bound is what the collision reduction of [13] yields at multiplicity two (Proposition 6.1, stated for order $\lfloor n/2 \rfloor$ and hence valid here).

graph on $N = 2^n$ vertices is given by a circuit or an oracle, and the target order is $K = n/2$. Jain, Li, Robere, and Xun [13, Section III] explicitly raise the question of the query complexity of this relation in the deterministic, randomized, and quantum models. This paper answers the quantum question up to a polynomial gap and clarifies the lower-bound record on both the quantum and the classical side.

The central observation is that the candidate sets of the constructive proof need not be materialized. After i rounds, membership in the current set is determined by the i adjacency constraints imposed by the previous pivots. These short predicates allow quantum search to sample from the same recursion at substantially smaller query cost. The threshold 4^{K-1} is the natural scale for this recursive construction.

1.1 Our results

We work in the promised valid-graph adjacency-oracle model. One coherent query returns the colour of a requested unordered vertex pair, and the output is a set of vertices inducing one colour. Our main result is the following.

▶ **Theorem 1.1** (Implicit-majority quantum Ramsey search). *Let $K \geq 2$, $N \geq 4^{K-1}$, and $0 < \eta < 1/2$. Given coherent edge-oracle access to a simple undirected graph on N vertices, there is a quantum algorithm which, with probability at least $1 - \eta$, outputs a K-clique or a K-vertex independent set using*

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

edge queries in the worst case. The algorithm verifies every returned witness and may output $\perp$ on the exceptional event.

At the succinct Ramsey scale, the theorem gives the following form.

▶ **Corollary 1.2.** *Let $n \geq 2$, $N = 2^n$, and $0 < \eta < 1/2$. Given coherent edge-oracle access to a simple graph on N vertices, one can find a clique or independent set of* *order* $\lfloor n/2 \rfloor + 1$ *using*

$$O\left(\sqrt{N}\log N \log \frac{\log N}{\eta}\right)$$

edge queries in the worst case.

Proof. Set $K = \lfloor n/2 \rfloor + 1$. Then

$$4^{K-1} = 2^{2\lfloor n/2 \rfloor} \leq 2^n = N,$$

so Theorem 1.1 applies. Moreover, $2^K \leq 2\sqrt{N}$, $K = O(\log N)$, and $\log(K/\eta) = O(\log(\log N/\eta))$. Substituting these estimates into the bound of Theorem 1.1 proves the claim. ◀

To our knowledge, this is the first sublinear worst-case query algorithm for the Ramsey relation in any model. Table 1 places it among the known bounds at the Ramsey scale, including the further results of this paper, which we now describe.

An estimation-free recursion.

Sampling a uniform candidate and following the colour of its edge to the current pivot selects a colour class with probability proportional to its size. A one-dimensional survival recurrence shows that this size-biased recursion completes with probability greater than $1/2$ from 4^{K-1} vertices, which gives an $O(2^K K^2 \log K \log(1/\eta))$-query algorithm without any majority estimation (Proposition 5.1). The same recurrence extends to every fixed number of edge colours (Corollary C.1).

A randomized lower bound.

Every randomized algorithm needs $\Omega(M^{1-1/\sqrt{2}})$ queries to find a homogeneous K-set in a graph on $M = 4^{K-1}$ vertices, and the bound holds on the uniform distribution $G(M, 1/2)$ (Proposition 5.2). This improves the $M^{1/4}$ black-box bound of [12, 14]. The proof transports the random-Painter weight estimate of [5] to the query model; Appendix B gives a self-contained version of that estimate.

A quantum speedup on random graphs.

On $G(N, 1/2)$, a greedy quantum neighbourhood search finds a K-clique with $\widetilde{O}(N^{1/4})$ queries (Proposition 5.3). Together with the distributional form of the lower bound, this gives a polynomial separation between the quantum and randomized query complexities of the Ramsey relation on its natural hard distribution (Corollary 5.4). We are not aware of an earlier provable quantum speedup for this relation. In the worst case, no separation is claimed: the randomized complexity is only known to lie between $N^{1-1/\sqrt{2}}$ and N.

The quantum lower-bound record.

The paragraph of [13] that raises the query question also reports an $N^{1-o(1)}$ quantum lower bound at the default parameters, obtained by composing its collision reduction with the multicollision lower bound of [18]. Theorem 1.1 shows that no such bound can hold at these parameters. Section 6.3 aligns the two formulations and shows that the cited ingredients compose to an exponent of at most $1/12$, and at most $1/24$ under the parameter condition as printed, with equality at multiplicity two; the resulting $\Omega(N^{1/12})$ bound (Proposition 6.1) is, to our knowledge, the best known quantum lower bound. The main theorems of [13] do not depend on the remark.

1.2 Technical overview

Both algorithms represent the candidate set after round i by the conjunction of the adjacency constraints generated by the first i pivots. A membership test therefore uses $O(i)$ edge queries. Capped unknown-solution search samples from such a set without first enumerating its vertices; conditional on success, permutation symmetry makes the sample exactly uniform. The precise sampler guarantee and its proof appear in Lemma 2.2 and Appendix A.

The first recursion removes majority estimation entirely. At each pivot, it samples a uniform candidate and follows the colour of the sampled edge. A colour class is therefore selected with probability proportional to its size. The survival probability obeys a one-dimensional recurrence, and Lemma 4.1 shows that the recursion completes with probability greater than $1/2$ from 4^{K-1} vertices. This yields the estimation-free bound in Proposition 5.1.

The sharper recursion estimates a majority colour at each pivot. Sampling a deep candidate set is exponentially more expensive than sampling an early one, so imposing the same estimation accuracy at every level wastes queries. We instead distribute a fixed total error budget across the recursion. Early majorities receive higher accuracy, while later majorities tolerate larger error. A dyadic schedule keeps the cumulative loss in candidate-set size bounded and balances the work across levels. Conditional concentration, the resulting density invariant, and exact output correctness are proved in Lemmas 4.2–4.4; the query summation in Section 5 then proves Theorem 1.1.

The lower bound and the random-graph speedup both rest on the fact that the common neighbourhood of a small vertex set in $G(N, 1/2)$ has predictable size. For the lower bound, a martingale weight argument in the style of [5] shows that an adaptive algorithm with T queries exposes a monochromatic K-set with probability at most $2 \cdot 2^{-\binom{K}{2}+c(c-1)}(2T)^{K-c}$ for every $c \leq K/2$; optimizing c gives the exponent $2 - \sqrt{2}$. For the speedup, a union bound shows that every j-set has common neighbourhood of density about 2^{-j}, so greedy quantum search costs only $O(2^{j/2})$ predicate evaluations at depth j and the recursion has depth K rather than $2K - 3$.

Organization.

Section 2 defines the query model and the sampler, and Sections 3–5 present the algorithms, the proofs, the lower bound, and the random-graph speedup. Section 6 compares the result with prior work and with the reported lower bound, and Section 7 concludes with open questions. The appendices contain the sampler proof, the lower-bound and random-graph proofs, the multicolour proof, numerical illustrations, and the AI disclosure.

2 Oracle model and capped quantum sampling

Let $G\colon \binom{[N]}{2} \to \{0,1\}$ be a promised simple undirected graph, extended symmetrically to ordered pairs with $G(u,u) = 0$. Its coherent edge oracle acts as

$$O_G|u,v,b\rangle = |u,v,b \oplus G(u,v)\rangle.$$

Query complexity counts calls to O_G. Reversible operations on vertex labels are free, as usual in the query model, and coherent access to O_G is part of the input specification.

The same asymptotic bound applies to the standard locally verifiable TFNP relation with arbitrary Boolean-circuit encodings.

▶ **Corollary 2.1** (Locally verifiable arbitrary encoding). *The standard Ramsey TFNP relation that accepts either a local invalidity certificate or a locally verified homogeneous set has the same $O(2^K K \log(K/\eta))$ quantum-query upper bound as Theorem 1.1.*

Proof. Given a circuit $C(u,v)$, define the canonical simple graph

$$A(u,v) = \begin{cases} C(\min\{u,v\}, \max\{u,v\}), & u \neq v, \\ 0, & u = v. \end{cases}$$

One query to A uses $O(1)$ queries to C, including uncomputation. Run Theorem 1.1 on A and let W be the returned homogeneous set. Query the diagonal and both ordered entries induced by W, using $O(K^2)$ additional queries. A self-loop or an asymmetric pair is a local invalidity certificate. Otherwise C agrees with A on W, so W is a valid clique or independent-set witness. The verification cost is absorbed by the bound in Theorem 1.1. ◀

Our algorithms use the following consequence of unknown-solution quantum search [2]. We include its proof in Appendix A because the analysis needs both bounded termination and exact conditional uniformity.

▶ **Lemma 2.2** (Capped uniform marked-item search). *Let $T \subseteq [M]$ have density at least $\lambda > 0$, and suppose its membership predicate costs q edge queries. There is a quantum block which uses at most $O(q/\sqrt{\lambda})$ edge queries, succeeds with probability at least an absolute constant $p_0 > 0$, and, conditional on verified success, returns an exactly uniform element of T. The same predetermined cap makes the block terminate if the density promise is false or T is empty.*

Moreover, m independent uniform samples can be collected, except with probability δ, using a predetermined $O(m + \log(1/\delta))$ blocks whenever the density promise holds.

The batch guarantee avoids a separate $\log(1/\delta)$ overhead for each sample. Every block starts from the uniform state, uses fresh internal randomness, measures a candidate, and verifies membership. Conditional on success, the returned vertex is uniform on T; successful outputs from distinct blocks are independent.

For comparison, consider the explicit classical recursion on $M = 4^{K-1} = 2^{r+1}$ vertices, where $r = 2K - 3$. At level i, query all edges from a pivot to the other $s_i = M/2^i$ candidates. A majority colour contains at least $s_i/2$ of the $s_i - 1$ non-pivot vertices; retain exactly $s_i/2$ of them. Then

$$\sum_{i<r}(s_i - 1) < 2M,$$

and two vertices remain after r rounds. The quantum algorithms retain the entire selected colour class but access it only through its membership predicate.

3 Two implicit-set recursions

3.1 An estimation-free size-biased recursion

Set $M = 4^{K-1}$ and $r = 2K - 3$, restrict the input to its first M vertices, let $S_0 = [M]$, and fix any $v_0 \in S_0$. At step $i = 0, \ldots, r-1$, define $T_i = S_i \setminus \{v_i\}$. Apply the repeated capped search from Lemma 2.2, with the universal density lower bound $1/M$, to sample an exactly uniform $x_i \in T_i$; abort if the fixed cap is exhausted. Query

$$c_i = G(v_i, x_i),$$

and set

$$S_{i+1} = \{x \in T_i : G(v_i, x) = c_i\}, \qquad v_{i+1} = x_i.$$

After the final step, put $w = v_r$. Among the r labels c_i, choose a colour that appears at least $K - 1$ times. Return the corresponding pivots together with w, after directly verifying the induced edges.

The algorithm never materializes S_i. Its membership predicate is the conjunction of the edge constraints imposed by the earlier pivots, together with the exclusions of those pivots, and costs $O(i + 1)$ edge queries. Repeating each capped block $O(\log(2r))$ times reduces its conditional failure probability on a nonempty set to $O(1/r)$. The probability that all r implicit sets remain nonempty is analyzed in Lemma 4.1.

3.2 Scale-aware approximate majorities

The sharper recursion estimates a near-majority colour at each pivot. Fix $M = 4^{K-1}$ input vertices and set

$$r = 2K - 3, \qquad E = \frac{1}{16}.$$

For $0 \le i < r$, define

$$d_i = \left\lceil \frac{r - 1 - i}{6} \right\rceil, \qquad z_i = 2^{-d_i}, \qquad Z = \sum_{j=0}^{r-1} z_j,$$

and allocate

$$\varepsilon_i = E\frac{z_i}{Z}, \qquad a_i = \frac{1}{2} - \varepsilon_i. \tag{1}$$

Thus $\sum_i \varepsilon_i = E$. The allowable error doubles once every six levels, matching the increasing cost of sampling the deeper candidate sets.

Before round i, the recorded pivots $v_0, \dots, v_{i-1}$ and colours $c_0, \dots, c_{i-1}$ define the exact implicit set

$$S_i = \{u \in [M] \setminus \{v_0, \dots, v_{i-1}\} : G(v_j, u) = c_j \text{ for every } j < i\}. \tag{2}$$

Its reversible membership predicate uses $O(i)$ edge queries.

At round i, use Lemma 2.2 to obtain a verified pivot $v_i \in S_i$. Then collect, with replacement,

$$m_i = \left\lceil \frac{1}{2\varepsilon_i^2} \log \frac{4r}{\eta} \right\rceil \tag{3}$$

independent uniform samples from $S_i \setminus \{v_i\}$. Query their edges to v_i, and let $\widehat{p}_i$ be the observed fraction of colour 1. Set $c_i = 1$ when $\widehat{p}_i \geq 1/2$ and $c_i = 0$ otherwise. Appending this edge constraint to Equation (2) defines S_{i+1}.

After r rounds, one final capped search returns $w \in S_r$. Select a colour that occurs at least $K-1$ times among $c_0, \dots, c_{r-1}$, take any $K-1$ pivots carrying that colour, and add w. Query all $\binom{K}{2}$ induced edges and return the set only if this direct check confirms homogeneity.

Every search and sampling batch has a predetermined cap. Hence every execution terminates even if an inaccurate earlier estimate invalidates a later density promise. A failed search or batch returns $\perp$.

4 Correctness and survival of the nested sets

We first analyze the estimation-free recursion. For integers $d, s \geq 0$, let $P_d(s)$ be the infimum, over all graphs, candidate sets of size s, and distinguished pivots in those sets, of the probability that the ideal size-biased recursion completes d additional steps. Set $P_0(s) = 1$ for $s \geq 1$, $P_d(0) = 0$, and $P_d(1) = 0$ for $d \geq 1$.

▶ **Lemma 4.1** (Size-biased survival). *For every $d \geq 0$ and $s \geq 1$,*

$$P_d(s) \geq \frac{(s - 2^d + 1)_+}{s}, \qquad (y)_+ = \max\{y, 0\}.$$

In particular, for $M = 4^{K-1}$ and $r = 2K - 3$, the ideal recursion completes with probability at least $1/2 + 1/M$.

Proof. The case $d = 0$ is immediate. Suppose $d \geq 1$ and $s \geq 2$, and let the two colour classes among the $s-1$ non-pivot vertices have sizes a and b. A uniform sample selects these classes with probabilities $a/(s-1)$ and $b/(s-1)$. The induction hypothesis therefore gives

$$P_d(s) \geq \frac{aP_{d-1}(a) + bP_{d-1}(b)}{s-1} \geq \frac{(a - 2^{d-1} + 1)_+ + (b - 2^{d-1} + 1)_+}{s-1}.$$

Because $(x)_+ + (y)_+ \geq (x+y)_+$ and $a+b = s-1$, the numerator is at least $(s-2^d+1)_+$. Replacing the denominator $s-1$ by s weakens the bound and proves the claim; the definitions cover $s \leq 1$. Finally, $M = 2^{r+1}$, so the bound at $(d,s) = (r,M)$ is $(M-2^r+1)/M = 1/2 + 1/M$. ◀

Exact conditional uniformity of each successful capped search couples an implemented run to the ideal recursion. Assign each of the r searches failure probability at most $1/(100r)$. A union bound decreases the completion probability by at most $1/100$, so a single fixed-cap run succeeds with probability at least 0.49. Independent repetition, followed by exact verification, amplifies this probability to $1-\eta$ using $O(\log(1/\eta))$ runs. On every completed run, nesting gives $G(v_i, v_j) = G(v_i, w) = c_i$ for $j > i$; the pigeonhole argument in Lemma 4.4 then gives a homogeneous output.

We next analyze the scale-aware recursion. The first lemma controls all adaptive estimates with one event.

▶ **Lemma 4.2** (Conditional near-majorities). *Except with probability at most $\eta/2$, every completed sampling batch reached after successful searches and an accurate prefix satisfies*

$$|\widehat{p}_i - p_i| \leq \varepsilon_i,$$

where p_i is the true colour-1 fraction from v_i to $S_i \setminus \{v_i\}$.

Proof. Condition on the complete classical history $\mathcal{H}_i$ before the samples at level i. The graph, S_i, and v_i are then fixed. Let $\mathcal{B}_i$ be the event that the verified batch succeeds. Conditional on $\mathcal{B}_i$, Lemma 2.2 gives independent uniform samples from the fixed set. Hoeffding's inequality [11] and Equation (3) imply

$$\Pr(|\widehat{p}_i - p_i| > \varepsilon_i \mid \mathcal{H}_i, \mathcal{B}_i) \leq 2e^{-2m_i\varepsilon_i^2} \leq \frac{\eta}{2r}.$$

Multiplying by $\Pr(\mathcal{B}_i \mid \mathcal{H}_i) \leq 1$ gives the same upper bound for a completed but inaccurate batch conditional on $\mathcal{H}_i$. Apply this bound to the first inaccurate completed level after an accurate prefix. The tower property removes the conditioning, and a union bound over the r possible levels completes the proof. The argument does not require independence across levels. ◀

There are r pivot searches, r sampling batches, and one final search. Give each operation failure probability at most $\eta/(4r+2)$ using the capped amplification in Lemma 2.2. Applying these conditional bounds after good prefixes, as above, shows that their union has probability at most $\eta/2$. Together with Lemma 4.2, all searches and estimates used below are good with probability at least $1-\eta$.

Let $s_i = |S_i|$ and

$$A_0 = 1, \qquad A_i = \prod_{j=0}^{i-1} a_j.$$

▶ **Lemma 4.3** (Variable recurrence and density). *On every prefix satisfying Lemma 4.2,*

$$s_i > MA_i - 1, \qquad A_i \geq \frac{7}{8}2^{-i}.$$

Consequently S_r is nonempty, and for every sampling level $i < r$,

$$\frac{|S_i \setminus \{v_i\}|}{M} > \frac{3}{8}2^{-i}.$$

The final set also satisfies $|S_r|/M \geq 2^{-(r+1)} \geq (3/8)2^{-r}$.

Proof. An accurate empirical majority retains at least the fraction $a_i = 1/2 - \varepsilon_i$ of the non-pivot vertices. Hence

$$s_{i+1} \geq a_i(s_i - 1). \tag{4}$$

Since $\sum_i \varepsilon_i = 1/16$ and $\prod_j(1 - x_j) \geq 1 - \sum_j x_j$ for $x_j \in [0, 1]$,

$$A_i = 2^{-i}\prod_{j<i}(1 - 2\varepsilon_j) \geq 2^{-i}\left(1 - 2\sum_{j<i}\varepsilon_j\right) \geq \frac{7}{8}2^{-i}.$$

We prove $s_i > MA_i - 1$ by induction. The claim holds at $i = 0$. If it holds at level i, then Equation (4) and $a_i < 1/2$ give

$$s_{i+1} > a_i(MA_i - 2) = MA_{i+1} - 2a_i > MA_{i+1} - 1.$$

Because $M2^{-r} = 2$, we obtain $s_r > (7/8)2 - 1 = 3/4$, and therefore the integer s_r is positive. For $i < r$, we have $M2^{-i} \geq 4$ and

$$s_i - 1 > MA_i - 2 \geq \frac{7}{8}M2^{-i} - 2 \geq \frac{3}{8}M2^{-i}.$$

Finally, $s_r \geq 1$ and $1/M = 2^{-(r+1)}$, which proves the last assertion. ◀

▶ **Lemma 4.4** (Exact homogeneous output). *Whenever the algorithm reaches its final verified output, that output is a K-clique or a K-vertex independent set.*

Proof. For $i < j$, the definition of S_{i+1} gives $G(v_i, v_j) = c_i$, and it also gives $G(v_i, w) = c_i$. Among the $r = 2K - 3$ colours, one occurs at least $\lceil r/2 \rceil = K - 1$ times. The corresponding pivots together with w therefore induce that colour. The final direct check verifies the returned set independently of earlier search or estimation errors. ◀

5 Query complexity, a classical lower bound, and random graphs

5.1 The estimation-free recursion

▶ **Proposition 5.1** (Size-biased quantum recursion). *Under the hypotheses of Theorem 1.1, the estimation-free algorithm returns a verified homogeneous K-set with probability at least $1 - \eta$ using*

$$O\left(K^2 2^K \log K \log \frac{1}{\eta}\right)$$

edge queries in the worst case.

Proof. A capped sample from a nonempty subset of the M-element universe costs $O(\sqrt{M})$ membership tests. Repeating the block $O(\log(2r))$ times reduces its failure probability to $O(1/r)$. Since the membership predicate at level i costs $O(i+1)$ edge queries, one fixed-cap run uses

$$O\left(\sqrt{M}\log(2r)\sum_{i=0}^{r-1}(i+1)\right) = O(K^2 2^K \log K)$$

edge queries. The survival analysis following Lemma 4.1 gives a constant success probability for each run, including capped-search failures and final verification. Repeating independent runs $O(\log(1/\eta))$ times proves the stated worst-case bound. ◀

5.2 The scale-aware recursion

Proof of Theorem 1.1. Set $M = 4^{K-1}$ and run the algorithm of Section 3 on any fixed set of M input vertices. The failure allocation in Section 4, together with Lemma 4.2, implies that all required searches succeed and all completed estimates are accurate with probability at least $1-\eta$. On this event, Lemma 4.3 guarantees every density promise used by the sampler and ensures that S_r is nonempty. The final output is homogeneous by Lemma 4.4. Because the algorithm verifies the returned set directly, every non-$\perp$ output is valid even outside the good event.

It remains to bound the number of queries. By Lemma 4.3, a capped search block at level i uses $O(2^{i/2})$ membership tests. Computing and uncomputing membership and querying the sampled edge costs $O(i+1)$ edge queries. Define

$$b_i = (i+1)2^{i/2}.$$

The batch cap in Lemma 2.2 uses $O(m_i + \log(r/\eta)) = O(m_i)$ blocks. By Equation (3), the total cost of the sampling batches is therefore

$$O\left(\log\frac{r}{\eta}\sum_{i=0}^{r-1} b_i\varepsilon_i^{-2}\right). \tag{5}$$

At most six indices share any value of d_i, and hence

$$Z = \sum_i 2^{-d_i} < 6\sum_{d\geq 0} 2^{-d} = 12.$$

Using Equation (1),

$$\sum_{i=0}^{r-1} b_i\varepsilon_i^{-2} = O\left(\sum_{i=0}^{r-1}(i+1)2^{i/2}4^{d_i}\right).$$

Write $h = r-1-i$. Since $4^{\lceil h/6\rceil} \leq 4\cdot 2^{h/3}$, the last display is at most

$$O\left(2^{r/2}\sum_{h=0}^{r-1}(r-h)2^{-h/6}\right) = O(r2^{r/2}).$$

The amplified pivot searches and the final search cost

$$O\left(\log \frac{r}{\eta} \sum_{i=0}^{r} b_i\right) = O\left(r 2^{r/2} \log \frac{r}{\eta}\right),$$

and the $O(K^2)$ output verification is smaller. Substituting $r = 2K - 3$ into Equation (5) gives

$$O\left(2^K K \log \frac{K}{\eta}\right),$$

as claimed. ◀

Each search begins in the uniform state on the fixed power-of-two universe $[M]$. The recorded pivot labels and colours define the reversible membership predicate for S_i, so the algorithm requires no materialized representation of any candidate set.

5.3 A randomized classical lower bound

We complement the upper bound with a randomized classical lower bound. It is the random-Painter estimate of Conlon, Fox, Grinshpun, and He [5, proof of Theorem 1] for online Ramsey numbers, transported to the query model. Because that estimate is a statement about uniformly random colourings, the bound holds on the uniform distribution $G(N, 1/2)$ and not only in the worst case. Appendix B gives a self-contained proof.

▶ **Proposition 5.2** (Randomized classical lower bound). *Let $K \geq 8$ and let $T = \lfloor 2^{(2-\sqrt{2})K-2} \rfloor$. For every $N \geq K$, every randomized algorithm that makes at most T edge queries outputs a homogeneous K-set of $G \sim G(N, 1/2)$ with probability at most $5/8$. Consequently, for $M = 4^{K-1}$, every randomized edge-query algorithm that finds a homogeneous K-set with worst-case probability at least $2/3$ uses*

$$\Omega\left(2^{(2-\sqrt{2})K}\right) = \Omega\left(M^{1-1/\sqrt{2}}\right)$$

queries.

The exponent $1 - 1/\sqrt{2} \approx 0.2929$ improves the $M^{1/4}$ black-box lower bound of [12, 14]. Jain et al. [13, Section III] already point to [5] for the best known deterministic lower bound; Proposition 5.2 records the randomized and distributional form that the same analysis yields. It leaves the randomized worst-case complexity between $\Omega(M^{1-1/\sqrt{2}})$ and the $O(M)$ of the explicit recursion, and it does not separate Theorem 1.1 from every classical algorithm in the worst case.

5.4 A quantum speedup on random graphs

On the distribution $G(M, 1/2)$ of Proposition 5.2, a greedy quantum neighbourhood search is much cheaper than the worst-case algorithm of Theorem 1.1: the common neighbourhood of every small clique in a random graph has essentially its expected size, so no majority has to be estimated and the recursion has depth K rather than $2K - 3$.

▶ **Proposition 5.3** (Quantum greedy search on random graphs)**.** *Let $K \geq 2$, $M = 4^{K-1}$, and $0 < \eta < 1/2$, and let $G \sim G(M, 1/2)$. There is a quantum algorithm which makes*

$$O\Big(K 2^{K/2} \log \frac{K}{\eta}\Big)$$

edge queries on every input, never outputs a set that is not a K-clique, and outputs a K-clique with probability at least $1 - \eta - \pi_K$ over G and its internal randomness, where $\pi_K = 2M^{K-1} e^{-2^{K-2}/12}$ satisfies $\pi_K < e^{-100}$ for all $K \geq 14$.

The proof, given in Appendix B.3, shows that with probability at least $1 - \pi_K$ every j-set with $j < K$ has at least $M 2^{-j-1}$ common neighbours, and then runs the capped search of Lemma 2.2 with density parameter 2^{-j-1} at level j. Combined with the distributional form of Proposition 5.2, this gives a provable quantum speedup for Ramsey search on random graphs.

▶ **Corollary 5.4** (Separation on random graphs at the Ramsey scale)**.** *Let $n \geq 26$, $N = 2^n$, $K = \lfloor n/2 \rfloor + 1$, $0 < \eta < 1/2$, and $G \sim G(N, 1/2)$.*

1. *A quantum algorithm making $O(N^{1/4} \log N \log(\log N/\eta))$ edge queries outputs a homogeneous K-set of G with probability at least $1 - \eta - \pi_K$.*
2. *Every randomized algorithm making at most $\lfloor 2^{(2-\sqrt{2})K-2} \rfloor = \Omega(N^{1-1/\sqrt{2}})$ edge queries outputs a homogeneous K-set of G with probability at most $5/8$.*

Since $1/4 < 1 - 1/\sqrt{2}$, the bounded-error quantum query complexity of the Ramsey relation on $G(N, 1/2)$ is polynomially smaller than its bounded-error randomized query complexity.

Proof. The subgraph induced on the first $M = 4^{K-1} \leq N$ vertices is distributed as $G(M, 1/2)$, so the first part follows from Proposition 5.3, using $2^{K/2} \leq 2^{n/4+1/2}$ and $K = O(\log N)$. The second part is the distributional statement of Proposition 5.2, which applies because $K \geq 8$. For the final sentence, note that $n \geq 26$ gives $K \geq 14$, so $\pi_K < e^{-100}$; with $\eta = 1/4$ the quantum algorithm succeeds with probability greater than $2/3$, while $5/8 < 2/3$. ◀

▶ Remark 5.5 (Classical greedy search). On an expanding graph in the sense of Appendix B.3, the same recursion runs classically by rejection sampling: at level j, draw a uniform $u \in [M]$ and test its adjacency to the j current clique vertices with at most j queries. Each trial succeeds with probability at least 2^{-j-1}, so $\lceil 2^{j+1} \ln(K/\eta) \rceil$ trials find v_{j+1} except with probability η/K. Hence, for $G \sim G(M, 1/2)$, the recursion outputs a verified K-clique with probability at least $1 - \eta - \pi_K$ using $O(K 2^K \log(K/\eta))$ queries, which is $\widetilde{O}(N^{1/2})$ at the Ramsey scale. This is the classical entry of Table 1; quantum search lowers the cost of level j from 2^j to $2^{j/2}$ predicate evaluations.

The size-biased recursion also extends to every fixed number of edge colours. The statement and proof appear in Appendix C.1.

6 Related work and parameter comparison

6.1 Quantum algorithms for other Ramsey tasks

Several quantum approaches to Ramsey problems address a different input and output task from the one studied here. Work based on adiabatic computation, quantum annealing, or quantum counting searches for colourings that establish or test Ramsey-number bounds [9, 1, 24, 22, 23, 21]. In those formulations, the search space consists of labelled graph colourings. Our input is one fixed graph given by an adjacency oracle, and the goal is to find a homogeneous set guaranteed to exist in that graph.

Generic quantum algorithms for clique, independent set, and fixed-subgraph containment operate in a closer oracle model [6, 4, 17, 26]. Quantum backtracking supplies another general search primitive [19]. Our algorithms additionally exploit the nested majority structure of the constructive Ramsey proof: each candidate set is represented by the constraints accumulated along the recursion and sampled as an implicit set. We are not aware of an earlier implementation of the constructive Erdős–Szekeres recursion by this mechanism in the adjacency-query model.

6.2 Constructive Ramsey search

Ramsey search also appears in proof complexity and TFNP. It is connected to weak and structured pigeonhole principles [15, 16], its white-box hardness follows from collision-resistant hashing [14], and it belongs to the polynomial long-choice class PLC [20]. The closest matching query relation is studied in [13], which shows that it is not black-box reducible to PIGEON and, in the paragraph "Query complexity of RAMSEY" of its Section III, raises the question of its query complexity in the deterministic, randomized, and quantum models. On the classical side, the online Ramsey game (see [5] and the references therein), in which Builder queries edges and Painter answers adversarially, is exactly the deterministic version of this question with an unbounded vertex set; the best known lower bound for it is the random-Painter bound of [5] used in Proposition 5.2.

6.3 Comparison with a reported quantum lower bound

Model and parameter alignment.

Definition II.6 of [13] uses $N = 2^n$ vertices, and the convention following Definition II.7 sets the default target to $K = n/2$. The unnumbered "Query complexity of RAMSEY" paragraph in Section III (p. 415) reports an $N^{1-o(1)}$ quantum lower bound, based on its Lemma III.1 and the multicollision lower bound of [18].

Away from the diagonal, the graph produced by Lemma III.1 from a function h and a fixed graph A_0 is

$$A_h(u, v) = \mathbf{1}[h(u) = h(v)] \vee A_0(h(u), h(v)), \qquad A_h(u, u) = 0.$$

It is a symmetric simple graph. A coherent query uses $O(1)$ queries to the value oracle for h, including uncomputation of the two hash values. Thus the reduction lies within our valid-graph promise. Moreover, Corollary 2.1 gives the same upper bound for the locally verifiable arbitrary-circuit relation, and Corollary 1.2 returns $\lfloor n/2 \rfloor + 1$ vertices, from which one may discard a vertex to meet the default target. The graph promise, encoding, and output size therefore align in the two formulations.

Exponent obtained by direct substitution.

Let H denote the multicollision range size and G the number of vertices in the resulting Ramsey instance. For fixed multiplicity t, the sufficient condition in Theorem I.3 of [13] is $G \geq H^{4t}/4^t$. At the boundary $G = \Theta_t(H^{4t})$, the exponent in [18] is

$$\alpha_t = \frac{2^{t-1} - 1}{2^t - 1}.$$

The exponent transferred to G is therefore

$$\frac{\alpha_t}{4t} = \frac{2^{t-1} - 1}{4t(2^t - 1)} \leq \frac{1}{24}, \qquad t \geq 2.$$

Equality holds at $t = 2$. For $t \geq 3$, the inequality $\alpha_t < 1/2$ gives $\alpha_t/(4t) < 1/(8t) \leq 1/24$. Increasing G relative to H only reduces the transferred exponent. The multicollision theorem is stated for fixed t, so the cited result also does not support a growing-multiplicity limit.

The printed condition is not tight. The argument in the proof of Lemma III.1 only uses that a homogeneous set S of the product graph maps to at most $K_0 - 1$ values of h, where $K_0 = 2\log_2 H$ is the forbidden order in A_0; hence a homogeneous set of order $(t-1)(K_0-1)+1$ already forces a t-collision, the reduction only needs $G = \Theta_t(H^{4(t-1)})$, and the transferred exponent improves to $\alpha_t/(4(t-1))$, which is $1/12$ at $t = 2$ and, for $t \geq 3$, below $1/(8(t-1)) \leq 1/16$. Direct composition of the cited ingredients therefore yields an exponent at most $1/12$, rather than $1 - o(1)$. Theorem 1.1 rules out an $N^{1-o(1)}$ lower bound at these parameters: the default relation has quantum query complexity $O(\sqrt{N} \log N \log\log N)$. The remark is not used elsewhere in [13], and the black-box separation theorems of that paper are unaffected.

The lower bound that the reduction does give.

The case $t = 2$ of the composition is valid and yields, to our knowledge, the best known quantum lower bound. We state it for the target order $\lfloor n/2 \rfloor$; a lower bound for a smaller target applies a fortiori to every larger target, in particular to the default $K = n/2$ of [13] and to the order $\lfloor n/2 \rfloor + 1$ of Corollary 1.2.

▶ **Proposition 6.1** (Quantum lower bound from collision finding). *Let $N = 2^n$ with $n \geq 8$. Every quantum algorithm that, given coherent edge-oracle access to a simple graph on N vertices, outputs a clique or independent set of order $\lfloor n/2 \rfloor$ with probability at least $2/3$ on every input makes $\Omega(N^{1/12})$ edge queries.*

Proof. Let $m = \lfloor n/4 \rfloor \geq 2$ and $H = 2^m$. By Erdős's bound $R(k,k) > 2^{k/2}$ for $k \geq 3$ [7], there is a graph A_0 on $[H]$ with no clique or independent set of order $2m$; fix one and hardwire it, so that evaluating A_0 costs no queries. For $h\colon [N] \to [H]$ let A_h be the graph displayed above. It is a simple graph, and one coherent query to A_h costs $O(1)$ queries to h.

Every homogeneous set S of A_h with $|S| \geq 2m$ contains two vertices with the same h-value. Indeed, if S is independent, then its h-values are distinct, because equal values force an edge, and $h(S)$ is an independent set of A_0; hence $|S| = |h(S)| \leq 2m - 1$, which is impossible. So S is a clique, $h(S)$ is a clique of A_0, and $|h(S)| \leq 2m - 1 < |S|$.

Since $2m \leq \lfloor n/2 \rfloor$, a set of the order in the statement qualifies. Given a q-query algorithm $\mathcal{A}$ as in the statement, run it on A_h, evaluate h on the $\lfloor n/2 \rfloor$ returned vertices, and output two of them with equal h-value. This finds a collision of h with probability at least $2/3$ using $O(q+n)$ queries to h, for every h. Finding a collision in a uniformly random function $[N] \to [H]$ with $N \geq H$ requires $\Omega(H^{1/3})$ quantum queries [25, 18], so $q + O(n) = \Omega(2^{m/3})$. Finally $m \geq (n-3)/4$ gives $2^{m/3} \geq 2^{-1/4} N^{1/12}$, and $n = o(N^{1/12})$, so $q = \Omega(N^{1/12})$. ◀

Together with Corollary 1.2, the quantum query complexity of the default Ramsey relation on N vertices lies between $\Omega(N^{1/12})$ and $O(\sqrt{N} \log N \log\log N)$.

7 Discussion and open questions

The query improvement rests on two complementary ideas. First, the algorithm stores each candidate set as the conjunction of the adjacency constraints created by earlier pivots. This representation supports quantum sampling without enumerating the surviving vertices. Second, the scale-aware analysis allocates estimation error according to the cost of the recursion level. The resulting schedule keeps the cumulative loss in candidate-set size bounded while concentrating most of the sampling effort at the less expensive early levels. Table 1 summarizes the resulting landscape; we close with the questions it leaves open.

Randomized worst case.

The randomized query complexity of the Ramsey relation on $N = 2^n$ vertices lies between $\Omega(N^{1-1/\sqrt{2}})$ (Proposition 5.2) and the $O(N)$ of the explicit recursion. We do not know a randomized algorithm with $o(N)$ worst-case queries: classical rejection sampling from the implicit set at depth i costs 2^i membership tests, so the deep levels of the recursion cost $\Theta(N)$ classically, whereas amplitude amplification makes them cost $\Theta(\sqrt{N})$. A randomized lower bound of $N^{1/2+\Omega(1)}$ would separate the worst-case quantum and randomized complexities.

Quantum lower bounds.

By Proposition 6.1 and Corollary 1.2, the quantum query complexity lies between $\Omega(N^{1/12})$ and $O(\sqrt{N} \log N \log\log N)$. Is $\sqrt{N}$ optimal? The final levels of the recursion search sets of constant size inside a universe of size $\Theta(N)$, which costs $\Theta(\sqrt{N})$ for unstructured search, but these sets are defined by $O(\log N)$ adjacency constraints, and the optimality of Grover search does not apply to them.

Random graphs.

Corollary 5.4 separates the quantum and randomized complexities on $G(N, 1/2)$, but no nontrivial quantum lower bound specific to that distribution is known, so the quantum exponent $1/4$ may not be optimal. On the classical side, Conlon et al. [5] conjecture that the exponent $2-\sqrt{2}$ in the lower bound $2^{(2-\sqrt{2})K}$ is not tight for cliques of fixed order and that the truth is $2^{(2/3+o(1))K}$, which would correspond to $N^{1/3+o(1)}$ at the Ramsey scale; their branch-and-fill strategy may also admit a quantum version with a smaller exponent than $1/4$.

Extensions.

The implicit-set method may extend to off-diagonal and hypergraph Ramsey search. Quantum walks give sublinear-query algorithms for fixed sub-hypergraph containment [10]; a Ramsey algorithm would additionally need to exploit the totality and recursive structure of the problem. For multiple edge colours, Corollary C.1 gives a size-biased algorithm, while a scale-aware analogue may yield a sharper dependence on the recursion depth.

Scope of the evidence.

Appendix C.2 reports finite experiments illustrating the survival recurrence and the conditional uniformity of the sampler. They are sanity checks; the query bounds follow from the analytic proofs in Sections 2–5 and the appendices. The paper claims query bounds only: it makes no claim about gate complexity, hardware, or a worst-case separation from every randomized algorithm.

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A Proof of the capped-search lemma

Proof of Lemma 2.2. Begin in the uniform superposition on $[M]$ and apply the randomized-iteration unknown-solution search of [2]. When the marked density is at least λ, truncate the geometric iteration schedule once its maximum iteration count reaches a sufficiently large constant multiple of $1/\sqrt{\lambda}$. The cited analysis gives a success probability $p_0 > 0$, independent of M and λ, using $O(1/\sqrt{\lambda})$ calls to the membership predicate. Because the cap is fixed before any oracle outcome is observed, the block terminates for every predicate.

For each fixed number of Grover iterations, all marked basis states have the same amplitude. Randomizing the iteration count preserves this permutation symmetry. After measuring a candidate, evaluate the exact membership predicate and accept only a marked outcome. Conditional on acceptance, the output is therefore uniform on the marked set T. Independent blocks reinitialize all registers and random choices, so their accepted outputs are independent conditional on the block-success indicators.

To collect a batch, run

$$B = C\big(m + \log(1/\delta)\big)$$

independent capped blocks, where the absolute constant C is chosen from p_0. The number of accepted blocks stochastically dominates $\mathrm{Bin}(B, p_0)$. A Chernoff lower-tail bound [3] makes the probability of fewer than m acceptances at most δ. Conditional on any success pattern with at least m acceptances, the first m accepted outputs are independent and uniform on T. ◀

B Proofs for the classical lower bound and the random-graph speedup

B.1 A weight bound for adaptive queries to a random graph

We record the random-Painter estimate of Conlon, Fox, Grinshpun, and He [5, proof of Theorem 1] in the form used in this paper. We follow their weight argument, taking the matching number of the queried graph as the potential; removing the critical edge from a maximum matching then leaves a matching of the smaller set, which keeps the recursion closed.

Throughout this subsection $V \geq 2$, and $G \sim G(V, 1/2)$ assigns independent uniform colours in $\{0, 1\}$ to the pairs of $[V]$. A deterministic algorithm queries T distinct pairs adaptively. Let $e_1, \ldots, e_T$ be the queried pairs, let Q_t be the graph formed by $e_1, \ldots, e_t$, and let $\mathcal{F}_t$ be the σ-algebra generated by the first t answers. Because the algorithm is deterministic, e_{t+1} is an $\mathcal{F}_t$-measurable pair distinct from $e_1, \ldots, e_t$, and its colour is therefore a uniform bit independent of $\mathcal{F}_t$.

For $U \subseteq [V]$ and $0 \leq t \leq T$, let $e_t(U)$ be the number of pairs inside U among $e_1, \ldots, e_t$, and call U *consistent at time* t if all of these pairs have colour 1. Define

$$w(U, t) = \begin{cases} 2^{-\binom{|U|}{2} + e_t(U)}, & U \text{ is consistent at time } t, \\ 0, & \text{otherwise,} \end{cases}$$

the conditional probability, given $\mathcal{F}_t$, that U becomes a colour-1 clique when its remaining pairs are revealed. Let $\nu(U,t)$ be the matching number of $Q_t[U]$; it is nondecreasing in t and increases by at most one when a pair is added. For integers $k \geq 2c \geq 0$ set

$$w_{k,c}(t) = \sum_{|U|=k,\ \nu(U,t)\geq c} w(U,t).$$

▶ **Lemma B.1** (Weight bound). *For all integers $k \geq 2c \geq 0$,*

$$\mathbb{E}\, w_{k,c}(T) \leq 2^{-\binom{k}{2}+c(c-1)}\, V^{k-2c}\, T^c.$$

Proof. *Martingale property.* Fix U. If $e_{t+1} \not\subseteq U$, then $w(U,t+1) = w(U,t)$. If $e_{t+1} \subseteq U$, then with probability $1/2$ the new pair has colour 1 and $w(U,t+1) = 2w(U,t)$, and otherwise $w(U,t+1) = 0$. Hence $\mathbb{E}[w(U,t+1) \mid \mathcal{F}_t] = w(U,t)$, and in particular $\mathbb{E}\, w(U,T) = w(U,0) = 2^{-\binom{|U|}{2}}$. Summing over all k-sets gives the case $c = 0$:

$$\mathbb{E}\, w_{k,0}(T) = \binom{V}{k} 2^{-\binom{k}{2}} \leq V^k 2^{-\binom{k}{2}}.$$

Monotonicity. Since $\nu(U,\cdot)$ is nondecreasing, $w_{k,c}(t+1) \geq \sum_{\nu(U,t)\geq c} w(U,t+1)$, and each event $\{\nu(U,t) \geq c\}$ is $\mathcal{F}_t$-measurable. Taking conditional expectations gives $\mathbb{E}\, w_{k,c}(t) \leq \mathbb{E}\, w_{k,c}(t+1)$.

Critical times. Let $c \geq 1$. For each U let $\tau(U) = \min\{t : \nu(U,t) \geq c\}$, with $\tau(U) = \infty$ if $\nu(U,T) < c$; this is a stopping time with $\tau(U) \geq 1$. Since $\{\tau(U) = t\} \in \mathcal{F}_t$ and $w(U,\cdot)$ is a martingale,

$$\mathbb{E}\, w_{k,c}(T) = \sum_{|U|=k} \sum_{t=1}^{T} \mathbb{E}\big[w(U,T)\, \mathbf{1}[\tau(U) = t]\big] = \sum_{t=1}^{T} \mathbb{E}\, w^*_{k,c}(t),$$

where

$$w^*_{k,c}(t) = \sum_{|U|=k,\ \tau(U)=t} w(U,t).$$

Charging. Fix t and write $e_t = \{x,y\}$. Suppose $|U| = k$ and $\tau(U) = t$. Then $\nu(U,t-1) = c-1$, $\nu(U,t) = c$, and $e_t \subseteq U$. Moreover, every maximum matching of $Q_t[U]$ contains e_t, since a maximum matching avoiding e_t would be a matching of size c in $Q_{t-1}[U]$. Fix such a matching M, let $M' = M \setminus \{e_t\}$, let A be the set of $2(c-1)$ vertices covered by M', and let $U' = U \setminus \{x,y\}$. Then M' is a matching of $Q_{t-1}[U']$, so $\nu(U',t-1) \geq c-1$. We claim that at most $2(c-1)$ of the pairs between $\{x,y\}$ and U' belong to Q_{t-1}. Indeed, if $xa \in Q_{t-1}$ with $a \in U' \setminus A$, then $M' \cup \{xa\}$ is a matching of size c in $Q_{t-1}[U]$, a contradiction; the same holds for y. For each $ab \in M'$, the pairs xa and yb cannot both lie in Q_{t-1}, since $(M' \setminus \{ab\}) \cup \{xa, yb\}$ would again be a matching of size c in $Q_{t-1}[U]$; likewise xb and ya cannot both lie in Q_{t-1}. Hence at most two of xa, xb, ya, yb lie in Q_{t-1}, and the claim follows by summing over the $c-1$ edges of M'.

Consequently $e_t(U) \le e_{t-1}(U') + 1 + 2(c-1)$, and U is consistent at time t only if U' is consistent at time $t-1$ and e_t has colour 1. Using $\binom{k}{2} - \binom{k-2}{2} = 2k-3$, we obtain

$$w(U,t) \le 2^{-(2k-2c-2)}\, w(U',t-1) \quad \text{if } e_t \text{ has colour } 1,$$

and $w(U,t) = 0$ otherwise. The map $U \mapsto U'$ is injective because $U = U' \cup e_t$. Therefore

$$w^*_{k,c}(t) \le \mathbf{1}[e_t \text{ has colour } 1]\, 2^{-(2k-2c-2)}\, w_{k-2,c-1}(t-1).$$

The indicator is independent of $\mathcal{F}_{t-1}$ with mean $1/2$, while $w_{k-2,c-1}(t-1)$ is $\mathcal{F}_{t-1}$-measurable. Taking expectations and using monotonicity,

$$\mathbb{E}\, w^*_{k,c}(t) \le 2^{-(2k-2c-1)}\, \mathbb{E}\, w_{k-2,c-1}(T),$$

and summing over $t \le T$ gives

$$\mathbb{E}\, w_{k,c}(T) \le T\, 2^{-(2k-2c-1)}\, \mathbb{E}\, w_{k-2,c-1}(T).$$

Iteration. Applying the last inequality c times, the exponents $2(k-2i)-2(c-i)-1$ for $i = 0,\dots,c-1$ sum to $2kc - 3c^2$, and the case $c = 0$ gives

$$\mathbb{E}\, w_{k,c}(T) \le T^c\, 2^{-(2kc-3c^2)}\, V^{k-2c}\, 2^{-\binom{k-2c}{2}} = 2^{-\binom{k}{2}+c(c-1)}\, V^{k-2c}\, T^c,$$

because $2kc - 3c^2 + \binom{k-2c}{2} = \binom{k}{2} - c(c-1)$. ◀

▶ **Corollary B.2** (Fully exposed monochromatic sets). *Let $K \ge 2$, let $V \ge K$, let $T \ge 1$, and let a deterministic algorithm make at most T adaptive edge queries to $G \sim G(V,1/2)$. For every integer $0 \le c \le K/2$, the expected number of K-sets all of whose $\binom{K}{2}$ pairs have been queried and carry the same colour is at most*

$$2 \cdot 2^{-\binom{K}{2}+c(c-1)} (2T)^{K-c}.$$

Proof. Simulate the algorithm while renaming vertices in order of first appearance in a query, using a table from original to canonical labels. This does not change the joint distribution of the queried graph and its colours, because the colours of distinct pairs are independent and uniform; the renamed algorithm queries pairs inside $[V']$ with $V' = \min\{V, 2T\}$, and the number of fully exposed monochromatic K-sets is unchanged. After the renamed algorithm halts, append distinct unqueried pairs inside $[V']$ in a fixed order until exactly $T' = \min\{T, \binom{V'}{2}\}$ pairs have been queried; this can only increase that number. A fully exposed colour-1 K-set U satisfies $\nu(U,T') = \lfloor K/2 \rfloor \ge c$ and $w(U,T') = 1$, so the number of such sets is at most $w_{K,c}(T')$. Lemma B.1 with $V' \le 2T$ and $T' \le T$ bounds its expectation by $2^{-\binom{K}{2}+c(c-1)} (2T)^{K-2c} T^c \le 2^{-\binom{K}{2}+c(c-1)} (2T)^{K-c}$. The same bound holds for colour 0 by symmetry. ◀

B.2 Proof of the randomized lower bound

Proof of Proposition 5.2. Let $\beta = 2 - \sqrt{2}$ and $\alpha = 1 - 1/\sqrt{2}$, so that $\beta = 2\alpha$ and $\alpha^2 - 2\alpha + 1/2 = 0$. Set

$$T = \lfloor 2^{\beta K - 2} \rfloor, \qquad c = \lfloor \alpha K \rfloor = \alpha K + \epsilon, \qquad -1 < \epsilon \leq 0.$$

Since $\alpha < 1/2$, we have $0 \leq c \leq K/2$, and $K \geq 8$ gives $T \geq 1$.

Consider first a deterministic algorithm making at most T queries to $G \sim G(N, 1/2)$. By Corollary B.2 and Markov's inequality, the probability that the queried pairs contain a fully exposed monochromatic K-set is at most

$$2 \cdot 2^{-\binom{K}{2} + c(c-1)} (2T)^{K-c}. \tag{6}$$

Since $2T \leq 2^{\beta K - 1}$, the base-2 logarithm of Equation (6) is at most

$$1 - \binom{K}{2} + c(c-1) + (K-c)(\beta K - 1) = 1 - \frac{K}{2} + \epsilon^2,$$

where the identity follows by substituting $c = \alpha K + \epsilon$ and $\beta = 2\alpha$ and using $\alpha^2 - 2\alpha + 1/2 = 0$. For $K \geq 8$ this quantity is at most -2, so the event has probability at most $1/4$.

Now condition on the query transcript and suppose that the algorithm outputs a K-set whose internal pairs have not all been exposed in one colour. If the exposed internal pairs contain both colours, the output is not homogeneous. If they contain exactly one colour, at least one unexposed pair must match that colour, which has conditional probability at most $1/2$, because unexposed pairs are independent fair coins given the transcript. If no internal pair has been exposed, the success probability is $2^{1-\binom{K}{2}} \leq 1/2$ for $K \geq 3$. Thus the conditional success probability is at most $1/2$ outside the fully exposed event, and the success probability under $G(N, 1/2)$ is at most

$$\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8}.$$

A randomized algorithm making at most T queries is a distribution over such deterministic algorithms, so its success probability on $G \sim G(N, 1/2)$ is also at most $5/8$. Finally, an algorithm whose worst-case success probability on graphs with M vertices is at least $2/3 > 5/8$ must make more than T queries, and $M = 4^{K-1}$ gives

$$2^{\beta K} = \Theta\left(M^{\beta/2}\right) = \Theta\left(M^{1-1/\sqrt{2}}\right),$$

which proves the proposition. ◀

B.3 Proof of the random-graph speedup

Proof of Proposition 5.3. Let $M = 4^{K-1}$. For $W \subseteq [M]$ let

$$C(W) = \{u \in [M] \setminus W : \ G(w, u) = 1 \text{ for every } w \in W\}$$

be the common neighbourhood of W. Call G *expanding* if $|C(W)| \geq M2^{-j-1}$ for every $j \in \{1, \ldots, K-1\}$ and every j-set W; for $W = \emptyset$ the bound $|C(\emptyset)| = M$ holds trivially.

Random graphs are expanding. Fix $1 \leq j \leq K-1$ and a j-set W. Then $|C(W)| \sim \mathrm{Bin}(M-j, 2^{-j})$ with mean $\mu = (M-j)2^{-j}$. Since $M = 4^{K-1} \geq 4(K-1) \geq 4j$, we have $\mu \geq \frac{3}{4}M2^{-j}$, hence $M2^{-j-1} \leq \frac{2}{3}\mu$ and $\mu \geq \frac{3}{4}M2^{-(K-1)} = \frac{3}{2} \cdot 2^{K-2}$. The Chernoff bound $\Pr[X \leq (1-\gamma)\mu] \leq e^{-\gamma^2\mu/2}$ [3] with $\gamma = 1/3$ gives

$$\Pr\left[|C(W)| < M2^{-j-1}\right] \leq e^{-\mu/18} \leq e^{-2^{K-2}/12}.$$

A union bound over the at most $\sum_{j=1}^{K-1} M^j \leq 2M^{K-1}$ sets W shows that G fails to be expanding with probability at most $\pi_K = 2M^{K-1}e^{-2^{K-2}/12}$. For the numerical claim, $M = 4^{K-1}$ gives

$$\ln \pi_K = \left(1 + 2(K-1)^2\right)\ln 2 - \frac{2^{K-2}}{12},$$

which equals $339 \ln 2 - 4096/12 < -106$ at $K = 14$, and $\ln \pi_{K+1} - \ln \pi_K = 2(2K-1)\ln 2 - 2^{K-2}/12 < 0$ for all $K \geq 14$, since the subtracted term doubles at each step while the first term grows by $4 \ln 2$. Hence $\pi_K < e^{-100}$ for all $K \geq 14$.

The algorithm. Set $W_0 = \emptyset$. For $j = 0, \ldots, K-1$, given the clique $W_j = \{v_1, \ldots, v_j\}$, run the capped search of Lemma 2.2 on $[M]$ for the predicate "$u \notin W_j$ and $G(v_i, u) = 1$ for all $i \leq j$", with density parameter $\lambda_j = 2^{-j-1}$, repeating the block $O(\log(K/\eta))$ times so that its failure probability is at most η/K whenever the density promise holds. The predicate costs $O(j+1)$ edge queries, including uncomputation. Let v_{j+1} be the verified output and $W_{j+1} = W_j \cup \{v_{j+1}\}$. If any search fails, output $\perp$; otherwise output W_K.

Correctness and cost. Every accepted vertex was verified to be adjacent to all earlier vertices, so a non-$\perp$ output is a K-clique for every input graph. If G is expanding, then the search at level j has marked density at least λ_j, and a union bound over the K searches shows that the algorithm outputs W_K with probability at least $1 - \eta$. Hence it succeeds with probability at least $1 - \eta - \pi_K$ over G and its internal randomness. Each capped block at level j uses $O(2^{j/2})$ predicate evaluations, so the total number of edge queries is

$$O\left(\log \frac{K}{\eta} \sum_{j=0}^{K-1} (j+1)2^{j/2}\right) = O\left(K2^{K/2} \log \frac{K}{\eta}\right)$$

on every input, because every block has a predetermined cap. ◀

C Multicolour extension, numerical illustrations, and AI disclosure

C.1 A multicolour extension of size-biased survival

Suppose every edge has one of $q \geq 2$ colours, and define

$$r = q(K-2) + 1, \qquad B_0 = 0, \qquad B_d = 1 + q + \cdots + q^{d-1} = \frac{q^d - 1}{q - 1} \quad (d \geq 1).$$

Choose the power of two

$$L_q = 2^{\lceil \log_2(2B_r) \rceil},$$

so $2B_r \le L_q < 4B_r$.

▶ **Corollary C.1** (Multicolour size-biased Ramsey search). *For $q, K \ge 2$, every q-edge-coloured complete graph on at least L_q vertices admits a bounded-error quantum algorithm that outputs a monochromatic K-clique using*

$$O\left(r^2\sqrt{L_q}\log(2r)\log\frac{1}{\eta}\right)$$

colour-oracle queries in the worst case, where r, B_r, and L_q are as defined above.

Proof. Run the size-biased recursion of Section 3 for r rounds on the first L_q vertices, now following the colour of a uniformly sampled edge among q possibilities. Let $P_d^{(q)}(s)$ be the minimum probability that the ideal recursion completes d further rounds from a candidate set of size s and a fixed pivot. We claim that

$$P_d^{(q)}(s) \ge \frac{(s - B_d)_+}{s}$$

for all $d \ge 0$ and $s \ge 1$. The case $d = 0$ is immediate. For $d \ge 1$ and $s \ge 2$, let the q colour classes among the $s - 1$ non-pivot vertices have sizes $a_1, \ldots, a_q$. The induction hypothesis gives

$$P_d^{(q)}(s) \ge \frac{\sum_{j=1}^{q} a_j P_{d-1}^{(q)}(a_j)}{s-1} \ge \frac{\sum_{j=1}^{q}(a_j - B_{d-1})_+}{s-1}.$$

Since $\sum_j a_j = s - 1$ and $B_d = 1 + qB_{d-1}$, the numerator is at least $(s - B_d)_+$. This proves the claimed bound after weakening the denominator from $s - 1$ to s. The preceding display also gives the stronger bound $(s - B_d)_+/(s-1)$ whenever $d \ge 1$ and $s \ge 2$. Consequently, starting from $L_q \ge 2B_r$, the ideal recursion survives all r rounds with probability strictly greater than $1/2$.

A membership predicate at level i contains i colour constraints and costs $O(i + 1)$ oracle queries. Using the universal density bound $1/L_q$, one capped sample costs $O((i + 1)\sqrt{L_q})$ colour queries. Repeating the capped block $O(\log(2r))$ times makes each sampling failure probability $O(1/r)$, so one fixed-cap run costs

$$O\left(\sqrt{L_q}\log(2r)\sum_{i=0}^{r-1}(i+1)\right) = O\left(r^2\sqrt{L_q}\log(2r)\right).$$

The union bound preserves a positive constant success probability. Since $r = q(K - 2) + 1$, one of the q recorded colours occurs at least $\lceil r/q \rceil = K - 1$ times. The corresponding pivots and the final vertex form a monochromatic K-clique by the same nesting argument as in Lemma 4.4. Independent repetition and direct verification amplify the success probability to $1 - \eta$ at an additional $O(\log(1/\eta))$ factor. ◀

C.2 Small-instance numerical illustrations

Ideal-sampler recursion. We ran the scale-aware recursion at $K = 3$, $N = 16$, and $r = 3$ for 1,000 trials in each of five graph families. The error schedule was $(\varepsilon_0, \varepsilon_1, \varepsilon_2) = (1/64, 1/64, 1/32)$, with sample counts $m_i = \lceil \log(600)/(2\varepsilon_i^2) \rceil$, corresponding to a total estimation-failure budget of 0.01. Pivots were sampled uniformly, majority-estimation samples were independent with replacement, and ties selected colour 1. We replaced quantum search by ideal uniform sampling to isolate the combinatorial recursion.

The deterministic inputs were the complete graph, the empty graph, and the balanced complete bipartite graph $K_{8,8}$. For the random inputs, one $G(16, 1/2)$ graph was held fixed across all 1,000 trials, while the final family drew a fresh $G(16, 1/2)$ graph in each trial. Every returned triple was verified; exhaustion or a nonhomogeneous output counted as failure.

Graph family on 16 vertices	failed/trials	minimum final $\lvert S_3 \rvert$
complete	0/1000	13
empty	0/1000	13
balanced complete bipartite $K_{8,8}$	0/1000	6
one fixed $G(16, 1/2)$	0/1000	2
fresh $G(16, 1/2)$ per trial	0/1000	2

All 5,000 trials returned homogeneous triples. The final column records the smallest observed value of $|S_3|$ in each family.

Isolated Grover sampler. We evolved a 16-dimensional state vector initialized in the uniform superposition, using marked-set sizes $1, 2, 3, 4, 8, 15$. The measurement distribution was averaged over a uniformly chosen number of Grover iterations from zero through four. The probability of measuring a marked item ranged from 0.4026746749 to 0.5973253251. Conditional on a marked outcome, the largest deviation from the uniform distribution on marked items was zero to machine precision. This calculation isolates the permutation symmetry used in Lemma 2.2.

Finite split-tree survival. Minimizing the ideal size-biased survival probability over every admissible colour-class split gives exact finite-depth comparisons with the analytic bounds. For nine binary levels starting from 1,024 candidates, the minimum was approximately 0.5086135566, above the lower bound 513/1024. For four ternary levels starting from 128 candidates, the minimum was approximately 0.7155732457, above 88/127.

C.3 AI Disclosure

We used ChatGPT and Claude as auxiliary tools for literature search, brainstorming, proof checking, and building experiment scripts. The authors determined the research direction and presentations, evaluated and revised all AI-assisted material, and take full responsibility for the correctness, originality, and final content of the paper, including every proof and reference.